- Serum transaminases are significant markers for diagnosis of liver and heart diseases.
- Many non-essential amino acids are synthesized using transamination reactions.
- Deamination is classified into oxidative deamination and non-oxidative deamination.
- Oxidative Deamination is a process in which the alpha amino group of the amino acid is removed as ammonia, coupled with oxidation.
- Oxidative deamination involving the removal of amino group of glutamate as ammonia is the key reaction involved in delivery of amino groups to the liver for urea synthesis.
- L-amino acid oxidases and D-amino acid oxidases are also involved in oxidative deamination.
- Glutaminase and asparaginase remove the amino groups of glutamine and asparagine.
- Decarboxylation reactions are involved in the formation of biologically important amines.
- After the amino group is lost, the keto acids formed from the amino acids like pyruvate, alpha keto glutarate will enter into the TCA cycle and are utilized for energy purposes.
- Amino acids can be glucogenic and /or ketogenic.
- Urea is produced from ammonia with the help of five enzymes and two of these enzymes are present in the mitochondria, while, the rest are in the cytosol. Out of the two amino groups of the urea, one amino group enters the urea cycle as carbamoyl phosphate, and the other enters as aspartate.
- Niacin (pyridine 3 carboxylic acid) is formed from the amino acid tryptophan.
- Many specialized products are formed from tyrosine in different tissues. In the nervous system, tyrosine serves as the precursor for dopamine. In the adrenal glands, it is converted to stress hormones namely, epinephrine and nor-epinephrine. In the thyroid gland, tyrosine is iodinated and converted to Thyroid hormones namely tri-iodothyronine and tetraiodothyronine.

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